

R-STDP Project Outline

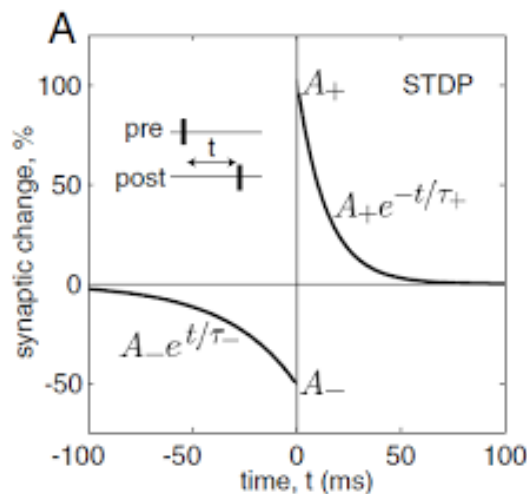
R-STDP Explained

Standard Spike Timing Dependent Plasticity (STDP)

- Describes a biologically plausible synaptic weight update rule that depends on pre and post synaptic activity.
- Hebbian learning = “Cells that fire together wire together”

$$\Delta t = t_{post} - t_{pre}$$
$$W(\Delta t) = \begin{cases} A_+ e^{-\Delta t / \tau_+}, & \text{if } \Delta t \geq 0 \\ -A_- e^{\Delta t / \tau_-}, & \text{if } \Delta t < 0 \end{cases}$$
$$\Delta w = \sum_{t_{pre}} \sum_{t_{post}} W(\Delta t)$$

(Bing et al., 2018)



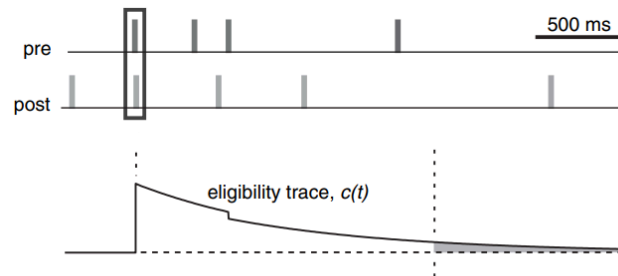
(Izhikevich & Desai, 2003)

- An **Eligibility Trace** for a synapse, in general, shows if there has been recent pre/post synaptic activity. Here is an example that uses an STDP rule across a time window:

$$\dot{c}(t) = -\frac{c}{\tau_c} + W(\Delta t) \delta(t - s_{pre/post}) C_1$$

(derivative of c) = $-c/\tau_c + W(\delta_t)\text{dirac_function}(t-s_{\text{pre/post}})C1$
 Where τ_c is the eligibility trace time constant and $C1$ is a scalar.

- Here is an example of an eligibility trace that uses a coincidence STDP rule (aka the eligibility trace is increased only when the pre and post synaptic neurons spike simultaneously):



The Role of Dopamine in Synaptic Plasticity, R-STDP

- Dopamine encodes reward prediction error (RPE)
 - Positive RPE → dopamine burst, learning that action is encouraged
 - Negative RPE → dopamine dip, learning that action is discouraged
 - 0 RPE → no change in dopamine, no learning / change in synapses
- Only synapses with recent STDP events are eligible for change
- The eligibility trace + DA combo links neural activity / actions to delayed consequences in the world

$$\dot{w}(t) = R(t) \times c(t)$$

(derivative of weight) = Reward function of time * $c(t)$

Hardware Implementation Ideas

Eligibility Trace = reuse the neuron membrane potential logic, remove v_{thresh} comparison since the eligibility trace doesn't spike.

STDP Rule = for now, for simplicity, just an AND gate somewhere. Not sure where to implement in the current framework.

Reward function = binary signal (either on for a number of timesteps or off for a number of timesteps for now. Can simplify it even further for now as just either being on or off per simulation as we're trying to figure out the rest of the implementation.

Problem to solve: Either (1) How to define RSTDP Synapses across memory banks and modules (has information on the pre and post synaptic spiking events per pair of pre/post neurons + each synapse gets its own eligibility trace) (2) how to add new logic modules that utilize existing memory framework and timestep framework.